

BORNFIRE

PROJECT DOCUMENTATION & PRODUCT SPECIFICATION



1. EXECUTIVE SUMMARY

Bornfire (formerly Focus Flow) is a next-generation productivity and social accountability web application designed to help users optimize their study routines, track daily positive habits, and engage in co-working sessions with their study squad.

At its core, Bornfire rejects standard clinical productivity layouts in favor of a bold, vibrant, gamified **Neo-Brutalist design system**. Incorporating elements reminiscent of social and gaming dashboards (e.g., Clash of Clans and PWA capabilities), it combines deep work tools (such as customized timers and task logs) with real-time squad tracking to introduce a friendly peer-accountability element.

2. TECHNICAL ARCHITECTURE

The system is constructed using a modern three-tier web application architecture designed to run reliably both locally and in cloud environments (such as Vercel).

- **Frontend Client:** Built using **React 18**, **TypeScript**, and **Vite**. Tailwind CSS is used for rapid styling under the customized Neo-Brutalist design specifications. Icons are rendered using Lucide React.
- **Backend & Database Layer:** Uses **Supabase** (PostgreSQL) for managed data storage and authentication. In local development environments, PostgreSQL runs within a Docker container (exposed on port 5435).
- **PWA capabilities:** Built using vite-plugin-pwa to enable standalone dashboard mode on desktop and mobile devices, improving accessibility.

3. COMPLETED WORK (FEATURES DONE)

The following features are fully implemented, tested, and pushed to the main repository branch:

FEATURE MODULE	IMPLEMENTATION DETAILS	STATUS
Authentication & User Profiles	Secure login/signup using Supabase. Generates a unique Bornfire ID (e.g., #BF-X7K9D2) using a custom character set to avoid visual confusion.	COMPLETED
Today Tasks & Focus Timer	Allows users to log tasks with priority levels (Low, Medium, High). Integrated Pomodoro focus timer updates remaining time and saves focused work duration directly.	COMPLETED
Habits Tracker	Tracks daily habit completion states with persistence across session reloads.	COMPLETED
Real-Time Squad Watch	Tracks squad members in real-time. Features a background heartbeat mechanism (every 30 seconds) and database pulling (every 15 seconds) to determine if a friend is online, actively working (running timer), or complete their daily task progress.	COMPLETED
Gamification & Leagues	Users earn XP by completing tasks and maintaining login streaks. The leaderboard displays user ranking, placing users in Bronze, Silver, Gold, Platinum, or Diamond leagues based on level.	COMPLETED
Branding & Logo Refinement	Rebranded application assets under "Bornfire". Re-rendered logo assets without overlapping text labels to serve as cleaner visual identifiers for mobile/desktop installs.	COMPLETED

4. PENDING / IN-PROGRESS WORK

The following items represent minor features or polish that are scheduled for completion in the next development cycle:

- **Profile Customization:** Allowing custom avatar uploads into Supabase Storage. Currently, avatars fall back to the initial character of the user's username.
- **Streak Freeze Shop:** A planned gamification store where users can spend points or XP to purchase streak freezes to protect their streak.
- **Detailed Task Analytics:** Historical progress graphs detailing weekly trends and total focus hours across categories in the Daily Progress module.

5. FUTURE PRODUCT ROADMAP

The long-term vision of Bornfire focuses on introducing direct peer connection and automated intelligent tools to assist users in preparing for placement exams and professional careers.

5.1 SQUAD VIDEO STREAMING (CO-WORKING ROOMS)

To deepen the peer accountability model, we will introduce co-working video streaming directly into the "Squad" section. Members can join visual study rooms alongside their friends.

- **Technical Approach:** Integrate WebRTC communication. We will leverage hosted APIs such as *Daily.co* or *Agora.io* to support low-latency video and audio tracks.
- **UI/UX Design:** Display small video cards next to each active friend's task progress, enabling a virtual library experience.

5.2 AI INTERVIEW & ENGLISH COACH BOT

An embedded AI mentor will be introduced to analyze squad chat transcripts and audio sessions. The bot will act as an evaluator to grade users on their professional communication and English speaking skills in normal life.

- **Technical Approach:** Utilizing the **Gemini 1.5 / 2.0 API**. User audio snippets will be sent to the Gemini API for translation, transcription, and grammatical analysis, providing customized feedback cards.
- **Features:**
 - Grades speaking/writing clarity.
 - Highlights filler words (e.g., "like", "uh", "um").
 - Provides tips to replace casual phrases with professional equivalents suitable for interviews.

5.3 AI PLACEMENT PREPARATION PLANNER

A text-to-schedule module that completely automates planning. Users can paste large chunks of text (such as raw syllabus lists, placement exam schedules, or company guidelines) and get a structured day-by-day plan.

- **Technical Approach:**

1. User pastes the preparation text into a planning text area.
2. The prompt is structured to send this text to the **Gemini API** utilizing *Structured JSON Outputs*.
3. The response returns an array of daily structured tasks mapped over a set timeline (e.g., 30 or 60 days).
4. The app parses this payload, stores it in the database under a new `placement_schedule` schema, and automatically populates the user's "Today" task list each morning.



STRATEGIC RECOMMENDATION

Implementing these future roadmap items will transform Bornfire from a personal task tracker into a comprehensive career preparation platform, increasing user engagement and utility.